Back **The Search for Exoplanets**
Graded Assignment • 30 min

Due Mar 20, 11:59 PM CET

1. Select the correct definition of an extrasolar planet (or exoplanet). 1 point

- ○ A planet orbiting a particularly bright star
- ◉ A planet residing beyond our Solar System
- ○ A planet orbiting two or more stars
- ○ A planet in other galaxy

2. Which feature is common to all main sequence stars? 1 point

- ◉ They are at a similar stage of evolution, burning hydrogen in their cores
- ○ They are all extremely luminous
- ○ All of them are likely to support potentially habitable planets
- ○ They all have a very similar temperature

3. Why is it difficult to detect new exoplanets directly? 1 point

- ○ We do not possess the technology for it
- ○ Directly detectable planets must pass in front of their parent star
- ○ Exoplanets are frequently veiled by dark matter
- ◉ Their light is often obscured by stellar radiation

4. What is the most commonly used method for discovering exoplanets? 1 point

- ○ The transit method
- ○ Astrometry
- ○ Gravitational lensing
- ◉ The Doppler shift (aka radial velocity) method

5. The transit method has enabled us to discover several new exoplanets. What else is it used for? 1 point

- ◉ Inferring physical properties of the planet
- ○ Tracking the movement of the parent star
- ○ Studying the influence of gravity on the 'bending' of light.

6. Why does the detection of exoplanets by gravitational lensing effectively rely on chance? 1 point

- ○ The planet-to-star mass ratio must be exactly right
- ◉ It requires the momentary alignment of two distant stars
- ○ The star in the foreground must 'wobble'

7. What is the Doppler shift (aka radial velocity) method based on? 1 point

- ◉ Detecting fluctuations in the wavelength of radiation received from a star
- ○ Detecting periodic reductions in stellar radiation
- ○ Tracking the position of a star over time

8. Which of the following statements is true? 1 point

- ○ Gravitational lensing is particularly useful for detecting planets near their parent stars
- ○ Planets near supergiants are particularly likely to be habitable
- ○ Astrometry relies on measuring Doppler shifts
- ◉ No currently used method for detecting exoplanets can be considered 'the best'

9. Exoplanets are incredibly diverse in terms of their environmental conditions. It is now clear that: 1 point

- ☐ Many exoplanets are entirely unlike those within our Solar System
- ☐ Most exoplanets are 'Hot Jupiters'
- ☑ Most exoplanets are uninhabitable
- ☐ The majority of exoplanets reside within 'habitable zones'

10. What proportion of known planetary systems is thought to have rocky planets within the habitable zone? 1 point

- ○ 25%
- ◉ 50%
- ○ 33%
- ○ 75%

11. Which of these are most likely to enable a planet to reside within the habitable zone? 1 point

- ☐ Orbiting a white dwarf
- ☐ A rocky surface
- ☑ Being within an optimal distance from the parent star
- ☐ Orbiting a main sequence star

12. Which two methods are the most conceptually similar? 1 point

- ○ The Doppler shift (aka radial velocity) method and the transit method
- ○ Gravitational lensing and the transit method
- ◉ Astrometry and the Doppler shift method

13. Which of the following statements about the habitable zone is false? 1 point

- ○ Its location is influenced by the luminosity of the parent star
- ◉ Cooler stars tend to have more distant habitable zones
- ○ The term 'habitable zone' is based on the likelihood of finding liquid water
- ○ Very little is known about the habitability of planets in binary star systems

14. Select the option that best describes the first definitively detected exoplanet. 1 point

- ○ It is a 'Hot Neptune'
- ○ It is small and rocky like the Earth, but it is too hot to be habitable
- ○ It is called Perseus
- ◉ The planet orbits close to its star and has a high mass

15. Astrometry relies on the change of what in a planetary system to detect a planet? 1 point

- ◉ The movement of a star around the common centre of mass
- ○ The heat of the star
- ○ The wavelength of light
- ○ The intensity of the light of the star

16. Most exoplanets described to date are very large. Tick the options that are most likely to explain this. 1 point

- ☑ There is a historical bias towards detecting large planets
- ☐ The majority of extrasolar planets are large
- ☑ We have only recently developed technologies for detecting small and Earth-like planets

17. What could be said about the habitable zones of hot stars? 1 point

- ○ Giant stars have the broadest habitable zones
- ○ The range of planetary orbits they may contain is unaffected by the star's luminosity
- ◉ They contain a comparatively broad range of possible planetary orbits that are further from their star

18. When a planet is part of a binary star system, this means that: 1 point

- ○ The planet is one of two orbiting the same parent star
- ◉ The planet is in a system with two stars
- ○ The radiation signal of the parent star alternates between two different modes

19. What are Michel Mayor and Didier Queloz recognized for? 1 point

- ◉ They discovered the exoplanet Bellerophon
- ○ They identified the first 'puffy' planet with a density the same as polystyrene
- ○ They speculated about the existence of exoplanets already in the 1500s
- ○ They were the first researchers to report the existence of an Earth-sized exoplanet

20. Name the most likely reason for the formation of 'diamond planets.' 1 point

- ◉ They orbit stars which are particularly rich in carbon
- ○ Tidal heating leads to the formation of vast layers of diamonds, graphite and silicon carbide
- ○ They have migrated extremely close to their parent stars